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18.06 Problem Set 3

+100*/100

3.2 Diff. Problem

18. $A = \begin{bmatrix} 0 & a & 0 \\ 0 & b & 0 \\ 0 & c & 0 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$

$x_1 + x_3 = 0$

$x_3 = 0$

$\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$ are independent

$x_1 + x_2 = b_1$

Diff. $x_2 + x_3 = b_2$ => no solution

36. $\begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix} \begin{bmatrix} x & y & z \\ w & v & t \end{bmatrix}$

$R_1 = \begin{bmatrix} a & b & c \\ 0 & e-b & f-c \end{bmatrix}$ $R_2 = \begin{bmatrix} x & y & z \\ 0 & v-y & t-z \end{bmatrix}$

$R_1 + R_2 = \begin{bmatrix} a+x & b+y & c+z \\ 0 & e-b+v-y & f-c+t-z \end{bmatrix}$

$A_1 + A_2 = \begin{bmatrix} a+x & b+y & c+z \\ d+w & e+v & f+t \end{bmatrix}$

The condition $rref(A_1+A_2) = R_1 + R_2$ $\not\Rightarrow A_1 = R_1;\ A_2 = R_2$ $\boxed{\text{False}}$

$R_1 + R_2 \neq rref(A_1 + A_2)$ since row reduction is not additive $\boxed{\text{False}}$

*All of Section 3.2 is different

24. a. A & A^T do not generally have the same nullspace as the free variables may change

$A = \begin{bmatrix} 1 & 0 & 4 & 3 \\ 3 & 1 & 2 & 5 \\ 0 & 0 & 1 & 7 \end{bmatrix}$ more free variables

=> $A^T = \begin{bmatrix} 1 & 3 & 0 \\ 0 & 1 & 0 \\ 4 & 2 & 1 \\ 3 & 5 & 7 \end{bmatrix}$ => more special sols.

$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 0 \\ 0 & 5 & 1 \\ 0 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ → Clean $rref(A^T)$

b. $A = \begin{bmatrix} 1 & 7 & 5 & 2 \\ 0 & 3 & 5 & 7 \\ 2 & 0 & 1 & 2 \end{bmatrix}$ free: x_4

$A^T = \begin{bmatrix} 1 & 0 & 2 \\ 7 & 3 & 0 \\ 5 & 5 & 1 \\ 2 & 7 & 2 \end{bmatrix}$ free: N/A

c. $A = \begin{bmatrix} 1 & 7 & 5 & 2 \\ 0 & 2 & 5 & 7 \\ 2 & 0 & 1 & 2 \end{bmatrix}$

=> $\begin{bmatrix} 1 & 7 & 5 & 2 \\ 0 & 2 & 5 & 7 \\ 0 & -14 & -9 & -2 \end{bmatrix} = \begin{bmatrix} 1 & 7 & 5 & 2 \\ 0 & 2 & 5 & 7 \\ 0 & 0 & 26 & 47 \end{bmatrix}$

$A^T = \begin{bmatrix} 1 & 0 & 2 \\ 7 & 2 & 0 \\ 5 & 5 & 1 \\ 2 & 7 & 2 \end{bmatrix}$ $= \begin{bmatrix} 1 & 7 & 5 & 2 \\ 0 & 1 & \frac{5}{2} & \frac{7}{2} \\ 0 & 0 & 1 & \frac{47}{26} \end{bmatrix}$ remains after rref(A) ...

Diff.

37. $A = \begin{bmatrix} 1 & 5 & 4 \\ 2 & 10 & 8 \\ 3 & 7 & 5 \end{bmatrix}$ $A^T = \begin{bmatrix} 1 & 2 & 3 \\ 5 & 10 & 7 \\ 4 & 8 & 5 \end{bmatrix}$

=> $\begin{bmatrix} 1 & 5 & 4 \\ 0 & 0 & 0 \\ 0 & -8 & -3 \end{bmatrix}$ => $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & -8 \\ 0 & 0 & -7 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & -8 \\ 0 & 0 & 0 \end{bmatrix}$

Rank = $\boxed{2}$ Rank = $\boxed{2}$

=> $\begin{bmatrix} 1 & 5 \\ 2 & 10 \end{bmatrix}$ will always lead to rank(A) = rank(A^T). For $n \times m$ matrix, the transpose always reciprocate the changes.

3.3 Diff.

14. rank(A)

$A = \begin{bmatrix} 1 & 1 & 5 \\ 0 & -1 & -4 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 5 \\ 0 & -1 & -4 \end{bmatrix}$

Rank = $\boxed{2}$

$A^T = \begin{bmatrix} 1 & 0 \\ 1 & -1 \\ 5 & -4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \\ 0 & -4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \\ 0 & 0 \end{bmatrix}$

Diff Rank = $\boxed{2}$

$A = \begin{bmatrix} 2 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 1 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$

Rank = $\boxed{2}$

$A^T = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix}$

Rank = $\boxed{2}$

Diff

$m \times n$

25. a. $r < m$, b. $r \leq m$, $r < n$, $r \leq n$

Diff. c. $r < m$, $r = n$ d. $r = m = n$

27. Exactly when A is a square matrix with $r = m = n$.

28. $[U\ \vec{0}] = \begin{bmatrix} 1 & 2 & 3 & 0 \\ 0 & 0 & 4 & 0 \end{bmatrix}$

$= \begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$

$= [R\vec{x}\ |\ \vec{0}]$

$[U\ \vec{c}] = \begin{bmatrix} 1 & 2 & 3 & 5 \\ 0 & 0 & 4 & 8 \end{bmatrix}$

$= \left[\begin{array}{ccc|c} 1 & 2 & 3 & 5 \\ 0 & 0 & 1 & 2 \end{array}\right] = \left[\begin{array}{ccc|c} 1 & 2 & 0 & -1 \\ 0 & 0 & 1 & 2 \end{array}\right]$

$= [R\vec{x}\ |\ \vec{d}]$

All of section 3.3 is also different!

~~[illegible]~~

Diff. 3.4

13. $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 3 & 1 \\ 3 & 1 & -1 \end{bmatrix}$ $U = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

a. $C(A) \Rightarrow \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & -2 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow$ Dimension of $\boxed{2}$

b. $C(U) = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ $\Rightarrow$ Dimension of $\boxed{2}$

c. $C(A^T) \Rightarrow \begin{bmatrix} 1 & 1 & 3 \\ 1 & 3 & 1 \\ 0 & 1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 3 \\ 0 & 2 & 0 \\ 0 & 1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 3 \\ 0 & 2 & 0 \\ 0 & 0 & -1 \end{bmatrix}$

Dimension of $\boxed{3}$

d. $C(U^T) \Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 1 & 2 & 0 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

Dimension of $\boxed{2}$

$\boxed{\text{C(A) and C(U) share the same space.}}$

Diff.

25. $A = \begin{bmatrix} 1 & 2 & 5 & 0 & 8 \\ 0 & 0 & c & 2 & 2 \\ 0 & 0 & 0 & d & 2 \end{bmatrix}$ $B = \begin{bmatrix} c & d \\ d & c \end{bmatrix}$

$\boxed{\text{for numbers } c = 0,\ d \in \mathbb{R} : d \neq 0 \text{ and } d = 0,\ c \in \mathbb{R} ; c \neq 0}$

Diff.

28. $\begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}$ with Enums $= 0$. Basis: $\boxed{\begin{bmatrix} a & a & -2a \\ -a & -a & 2a \end{bmatrix}}$ for $a \in \mathbb{R}$

A basis would be:

$\begin{bmatrix} a \\ a \end{bmatrix} + \begin{bmatrix} a \\ a \end{bmatrix} - 2\begin{bmatrix} a \\ a \end{bmatrix}$

$\Rightarrow \begin{bmatrix} a & a & -2a \\ a & a & -2a \end{bmatrix}, \begin{bmatrix} -2a & a & a \\ -2a & a & a \end{bmatrix}, \begin{bmatrix} a & -2a & a \\ a & -2a & a \end{bmatrix}$

for $a \in \mathbb{R}$

Diff.

35. $p(x) \leq 3$ $P(1) = 0$

$\Rightarrow \boxed{ax^3 + bx^2 + cx + d}$ $\boxed{ax^3 + bx^2 + cx + d - (ax^3 + bx^2 + cx + d)}$

for $\{(a, b, c, d) \in \mathbb{R}^4\}$

Diff.

36. $(a, b, c, d) := a + c + d = 0$

$\Rightarrow e + e + (-2e)$

$\therefore$ for all permutations of $\begin{bmatrix} e \\ e \\ -2e \end{bmatrix}$

$(a, b, c, d) := b + a = 0$

$\Rightarrow d - d$

$= \begin{bmatrix} d \\ -d \end{bmatrix}$ or $\begin{bmatrix} -d \\ d \end{bmatrix}$

$(a, b, c, d) := c = 2d$

$\Rightarrow \frac{c}{d} = 2$

$\therefore$ T has basis of

$= \boxed{\begin{bmatrix} e \\ -e \\ e \\ \frac{e}{2} \end{bmatrix}}$ for $e \in \mathbb{R}$

Dimension for S∩T is $\boxed{2}$ as there are 2 common elements.

* Since all problems within this pset are different, I resorted to grading it seperately. Therefore a more fitting guide is N/A; but that doesn't give the same depending...